Simulation Project 2

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Feb 1, 2026

Part 1

Question 1.

We will measure the empirical power of this test for difference of proportions as the proportion of trials where we have a significant result. Thus we will generate 1,000 experiments of 10,000 people for control and ad groups. Then find the proportion of each of those groups who purchased the product and perform a two-proportion z-test to determine if there is a difference in proportions. We will then count how many of those z-tests return a significant p-value and that will be our empirical power.

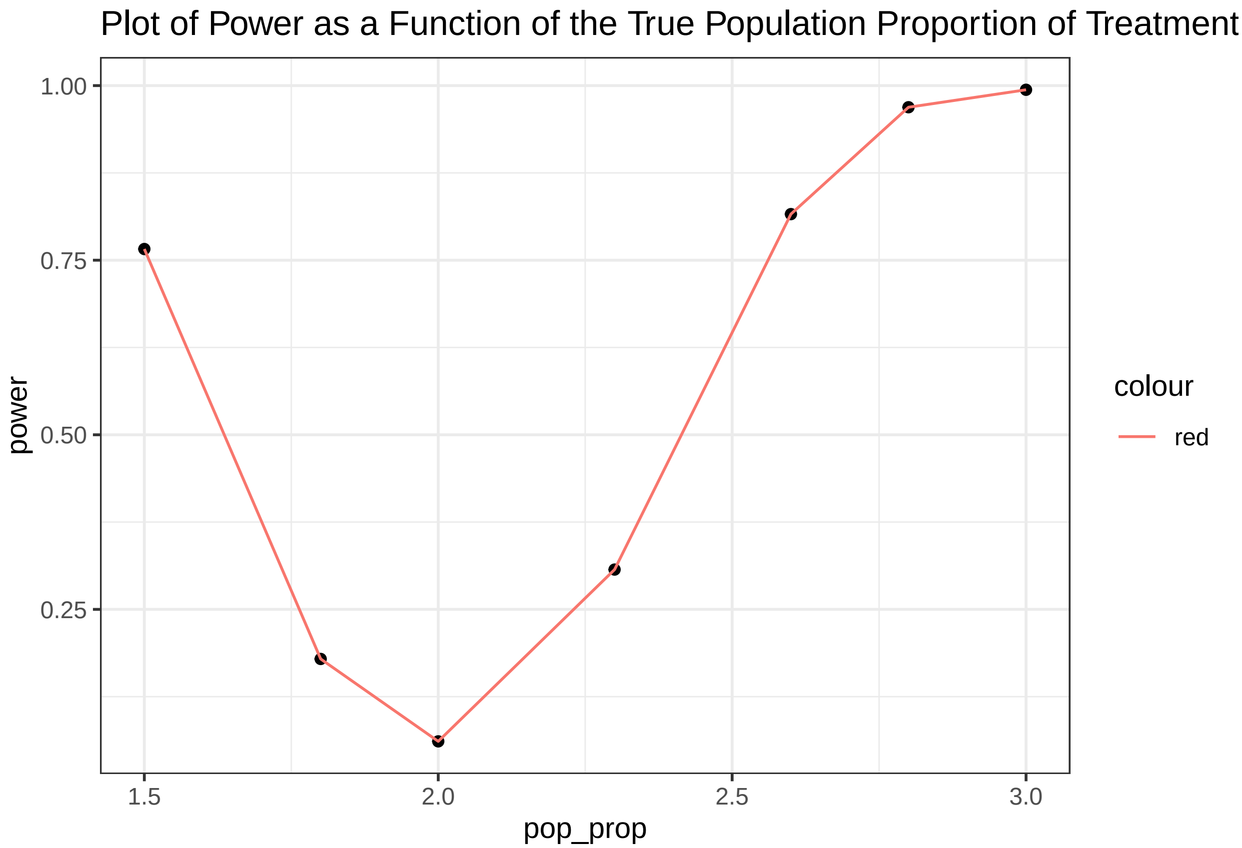
We found that out of our 1,000 tests we found 674 of them returned a p-value that was significant at the .05 level. Thus the empirical power of our two-proportion z-test is .674.

Question 2.

We will perform the same steps that we took in step two but with the changed sample sizes. What we find is that for scenario (1) we get an empirical power of .822. For scenario (2) we get an empirical power of .302. For our final scenario (3) we get an empirical power of .934. We can see that we have a greatly increased empirical power if we add the people to either both the control and the treatment groups or just the treatment group. But we see a great decrease in empirical power when we add the individuals to the control.

Question 3.

Next, we'll explore what happens to the power of the test as we change the true proportion of those who purchase the product in the treatment group. Conduct 1,000 simulations where the true proportion who purchase the product in the treatment group varies between 1.5% and 3% (use 1.5%, 3%, and several numbers in between). Continue using the original 10,000 in each group. Create a plot of the power of the test as a function of the true proportion of those who purchase the product in the treatment group.



We can see that as the true proportion of the treatment group gets closer to the true proportion of the control group the power greatly decreases. This makes sense because empirical power is the proportion of tests where we get a statistically significant result. We would not expect to get many statistically significant results when the true population proportions are the same because our significant result says that we have evidence to suggest that there are differences in the proportions.

Paragraph

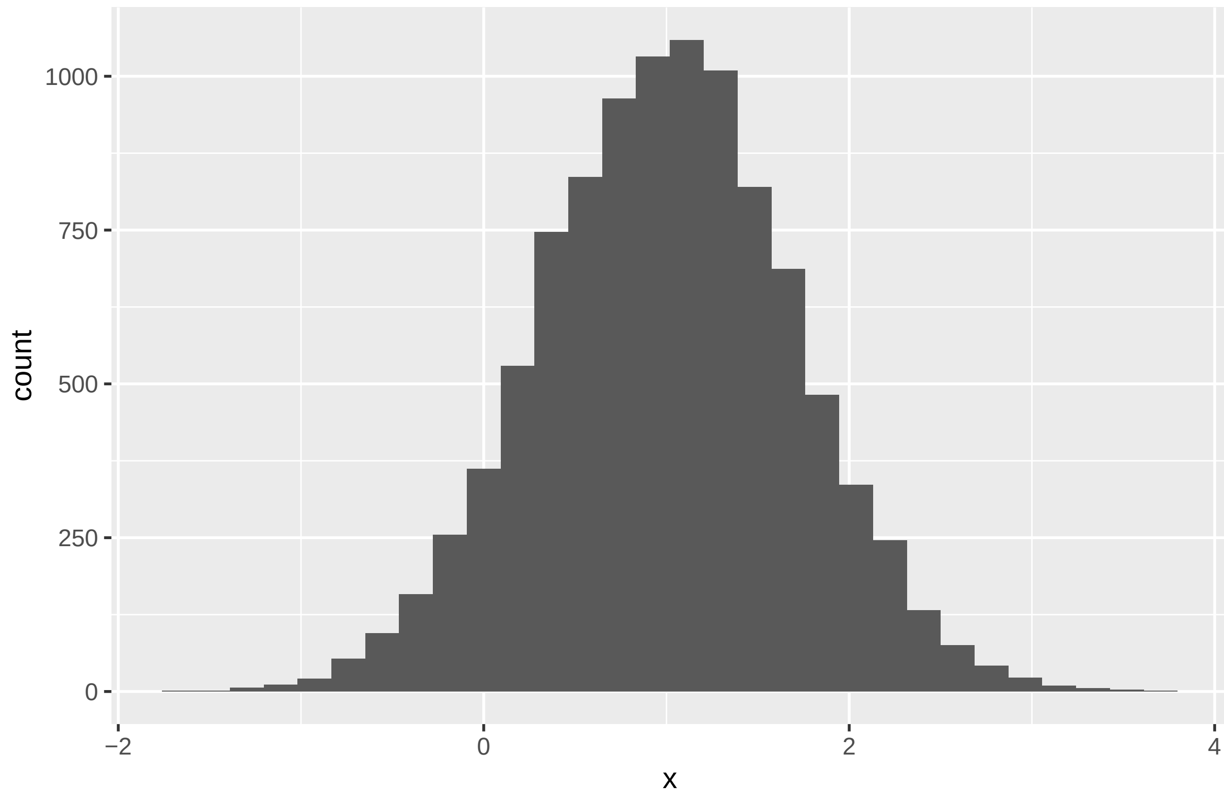
What we found is that there are a few factors that change the empirical power of a test. We found that the sample sizes and true population proportions can greatly effect the power. For sample size, increasing the number of participants in the control and treatment groups increases the power and increasing the number of participants in the treatment group only increases the power even more! We also found that increasing the sample size of the control group decreases the empirical power. As for the true population proportion, we found that as they get closer to each other (as $p_{control} = p_{treatment}$) the empirical power decreases. These findings relate to linear regression because linear regression with a binary predictor is the same as a two sample t-test assuming equal variances.

Implications for these findings in practice is the idea that we should design our experiments around these concepts to maximize our empirical power. It is also important because if we are given the sample sizes and the true population proportions we can get a good idea about how strong the empirical power will be without actually performing any tests.

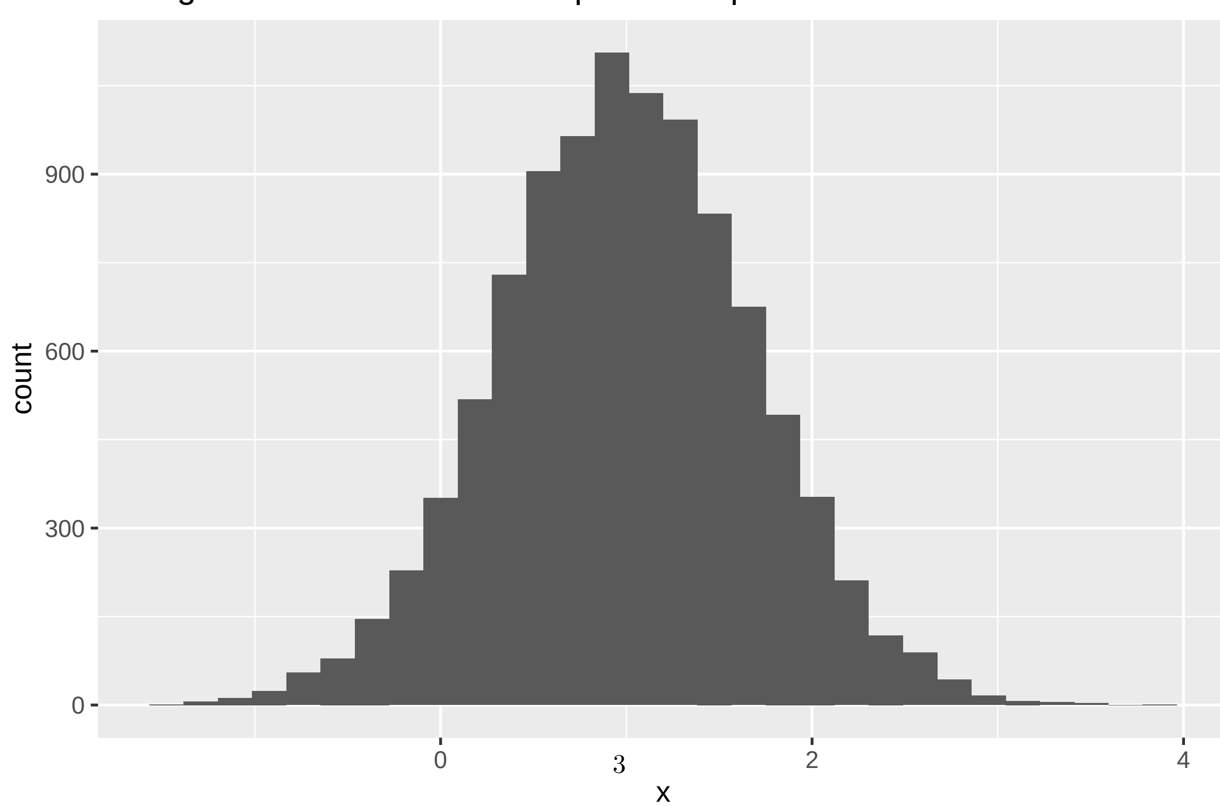
Part 2 - Normality

Question 1.

Histogram of Estimated Normal Slopes



Histogram of Estimated Chi−Squared Slopes



For the histogram of the normal distribution's β_1 values we can see it is centered slightly above 1, but is a nearly normal curve with a range of -2 to 4.

For the histogram of the chi-squared distribution's β_1 values we can see it is centered slightly below 1, has a somewhat normal curve, and has a range of -2 to 4.

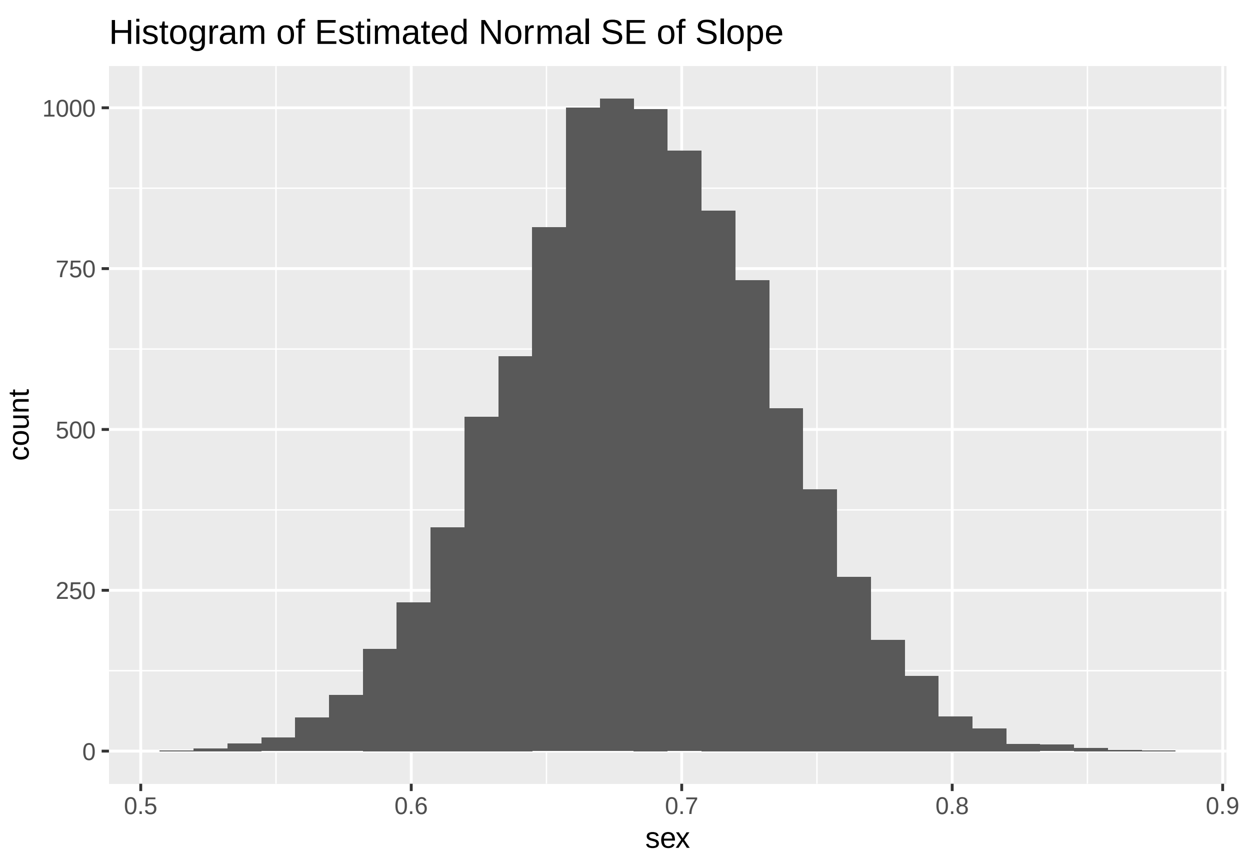
It is surprising to me that these are nearly the exact same plot! They are both centers around 1 and have the same range. They are also both close approximations of the normal curve. This is surprising because we are mapping random values between 0 and 1 to normal and chi-squared values. Based on graphs of these two distributions we can see that they are not very similar. So I am surprised that they have such similar plots of β_1.

Question 2.

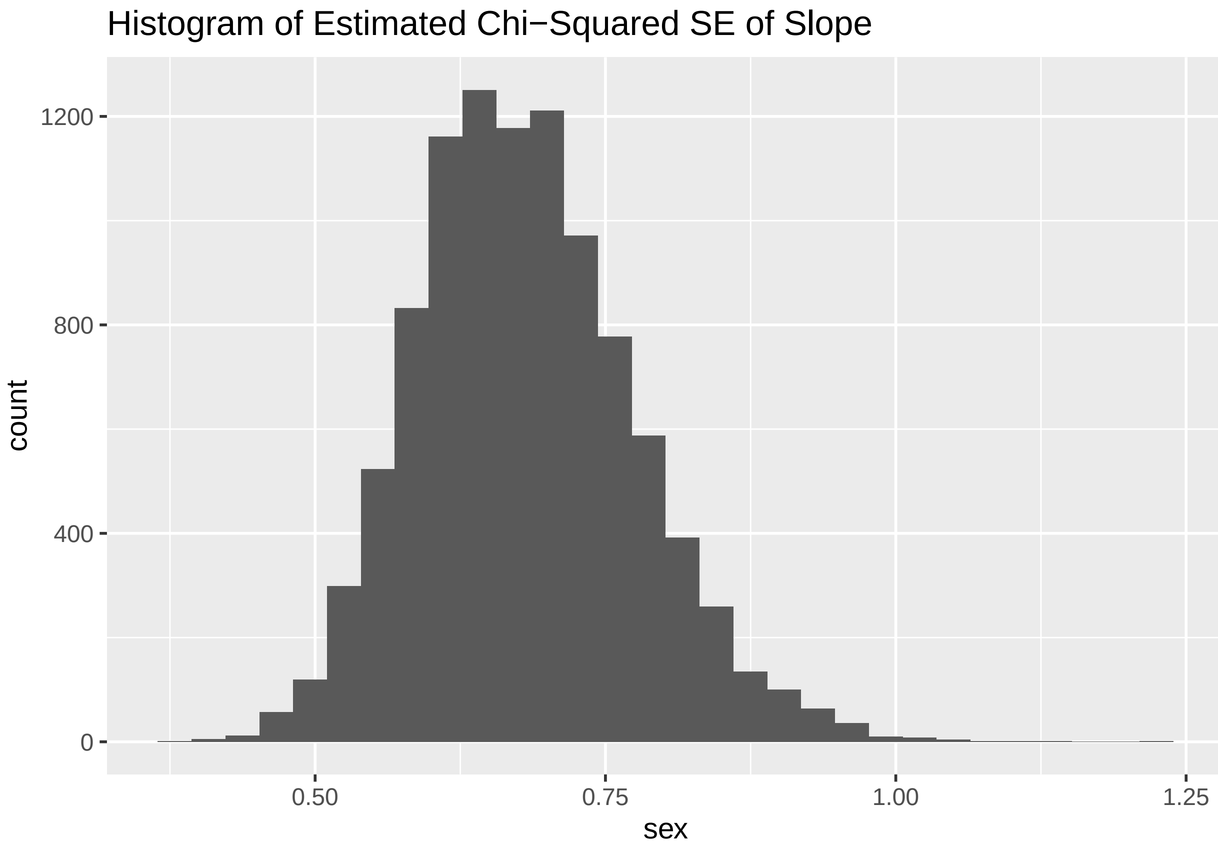
The theoretical stand errors of $\beta_1 = \sqrt{\sigma^2 * (1/\sum(x_i - \hat{x})^2)} = 0.6979592$ for the normal distribution and 0.6842273 for the Chi-Squared distribution.

We can now look at the histogram of the 10,000 estimated standard errors for the normal distribution and see that it is centered on .67 with a range between 0.5 and 0.9. It is a bell curve and looks close to normally distributed.

```
## ‘stat_bin()‘ using ‘bins = 30‘. Pick better value with ‘binwidth‘.
```




```
## ‘stat_bin()‘ using ‘bins = 30‘. Pick better value with ‘binwidth‘.
```



For the histogram of the 10,000 Chi-Squared SE's of the β_1 we can see that it is slightly right skewed, which makes sense since the Chi-Squared distribution is heavily right skewed. We can see that the mode is around .6 and has a range of 0.40 - 1.25.

It does make sense that each of our histograms are centered on the theoretical standard error for each distribution.

Question 3.

We know that the true mean for our normal distribution is 0, so to find the coverage probability of 95% confident intervals we check the proportion of the intervals that include our true mean of 0. We get a coverage probability of 0.9495.

Doing the same for the Chi-Squared distribution we know the true mean has been adjusted to 0. So we find the coverage probability of 95% confidence intervals to be .9463. What we can see is that the coverage probabilities are extremely close to each other. We also see that they are slightly lower than our true confidence interval of 95%.

Paragraph

The Chi-squared distribution is *very* different than the normal distribution; however, when we performed analysis on it in comparison to the normal distribution we did not see drastic differences in performance. We saw that the histograms for the estimated slopes and the histograms for the estimated standard error of the slope were pretty similar. We also did not notice any distinct changes in the coverage probabilities for confidence intervals. This is significant because normality is one of our assumptions for linear regression. But in practice, distributions are almost never perfectly normal. So our findings imply that the assumption of normality is not as necessary as we may have thought if we are performing confidence intervals or hypothesis tests on linear models.

Apendix

Part 1

Question 1

```r
library(tidyverse)
library(ggplot2)
# Make results reproducible
set.seed(5571)

#dataset
datasetyy <-data.frame(pvalue = NA)

## Example code to simulate control group.
## This gives us the number in the control group who purchase the product
##   out of 10,000 when truep = 0.02
#first group
for (x in 1:1000) {
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.025)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE 326
## 2    TRUE 674
```

```r
#We get 674 are less than 0! yay!
```

Question 2

```
#dataset
datasetyy <-data.frame(pvalue = NA)

#first group
for (x in 1:1000) {
num_gp1 <- rbinom(1, 15000, 0.02)
#second group
num_gp2 <- rbinom(1, 15000, 0.025)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE 178
## 2    TRUE 822
```

```
#count of 822!

#second situation!!!

datasetyy <-data.frame(pvalue = NA)

#first group
for (x in 1:1000) {
num_gp1 <- rbinom(1, 11000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.025)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE 724
## 2    TRUE 276
```

```
#get a count of 302

#final situation!!

datasetyy <-data.frame(pvalue = NA)

#first group
for (x in 1:1000) {
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 11000, 0.025)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE  52
## 2    TRUE 948
```

```
#get a count of 934
```

Question 3

```
#blank dataset
datasetyy <-data.frame(pvalue = NA)

for (x in 1:1000) {
#first group
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.015)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}
```

```r
#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE 218
## 2    TRUE 782
```

```r
#get a count of 766
#now adding that to our power df <3
powerdf <- data.frame(pop_prop = 1.5, power = .766)

#now repeating for a ton of true pop proportions!
datasetyy <-data.frame(pvalue = NA)

for (x in 1:1000) {
#first group
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.018)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE 800
## 2    TRUE 200
```

```r
# we get 179 which is crazy different?

#adding it to the powerdf
powerdf[nrow(powerdf) + 1,] = c(1.8, .179)

datasetyy <-data.frame(pvalue = NA)
```

```r
for (x in 1:1000) {
#first group
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.02)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE 940
## 2    TRUE  60
```

```r
# we get 179 which is crazy different?

#adding it to the powerdf
powerdf[nrow(powerdf) + 1,] = c(2.0, .061)


datasetyy <-data.frame(pvalue = NA)

for (x in 1:1000) {
#first group
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.023)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
```

```
## 1   FALSE 689
## 2    TRUE 311
```

```r
powerdf[nrow(powerdf) + 1,] = c(2.3, .307)


datasetyy <-data.frame(pvalue = NA)

for (x in 1:1000) {
#first group
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.026)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE 176
## 2    TRUE 824
```

```r
powerdf[nrow(powerdf) + 1,] = c(2.6, .816)



datasetyy <-data.frame(pvalue = NA)

for (x in 1:1000) {
#first group
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.028)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = F
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
```

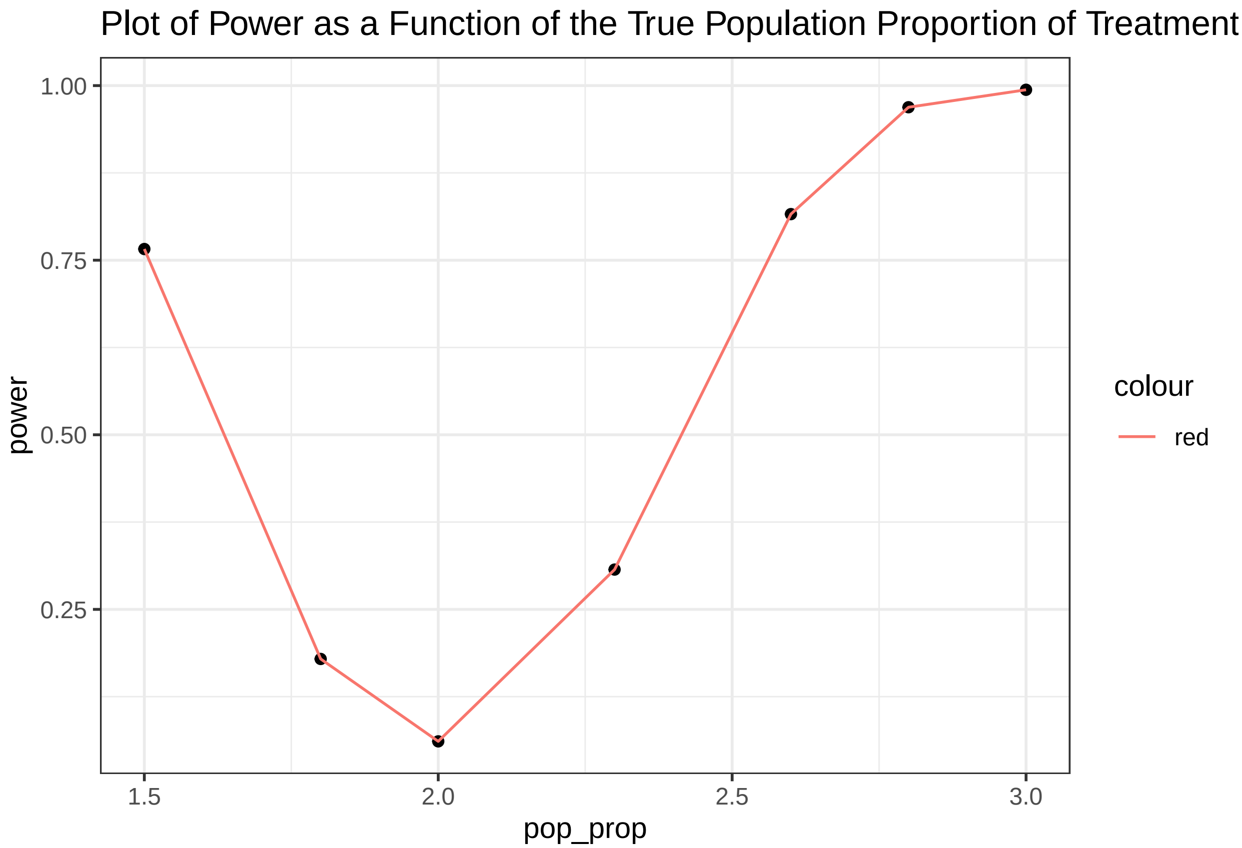
```r
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

```
##   yes < 0   n
## 1   FALSE  38
## 2    TRUE 962
```

```r
powerdf[nrow(powerdf) + 1 ,] = c(2.8, .969)

datasetyy <-data.frame(pvalue = NA)

for (x in 1:1000) {
#first group
num_gp1 <- rbinom(1, 10000, 0.02)
#second group
num_gp2 <- rbinom(1, 10000, 0.03)
#two proportion z - test
test = prop.test(x = c(num_gp1, num_gp2), n = c(10000, 10000), alternative = c("two.sided"), correct = 
datasetyy[nrow(datasetyy) + 1,] = test$p.value
}

#negative ones will be the ones that had a value smaller than .05
datasetyy2 <- datasetyy %>%
  mutate(yes = pvalue - .05)

#drop first row
datasetyy2 <- data.frame(datasetyy2[-1,])

datasetyy2 %>%
  count(yes < 0)
```

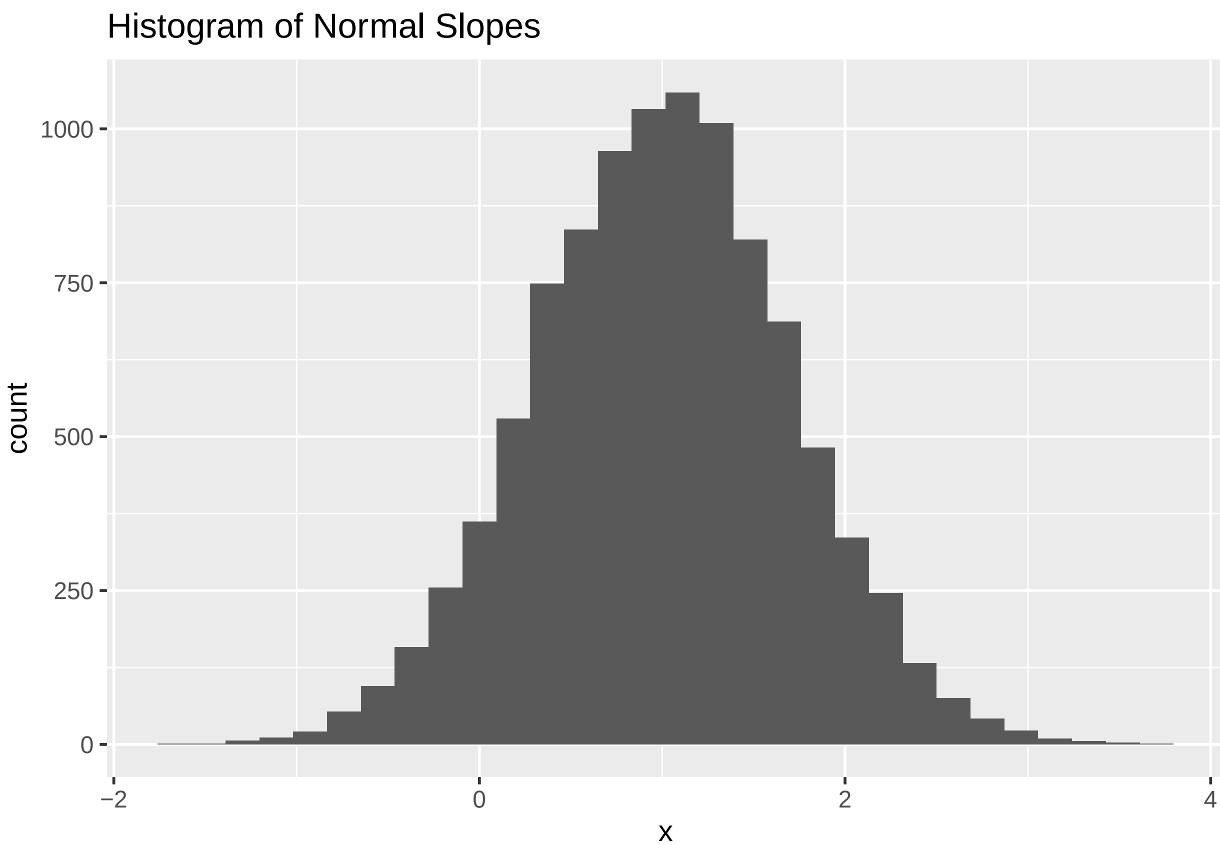
```
##   yes < 0   n
## 1   FALSE   6
## 2    TRUE 994
```

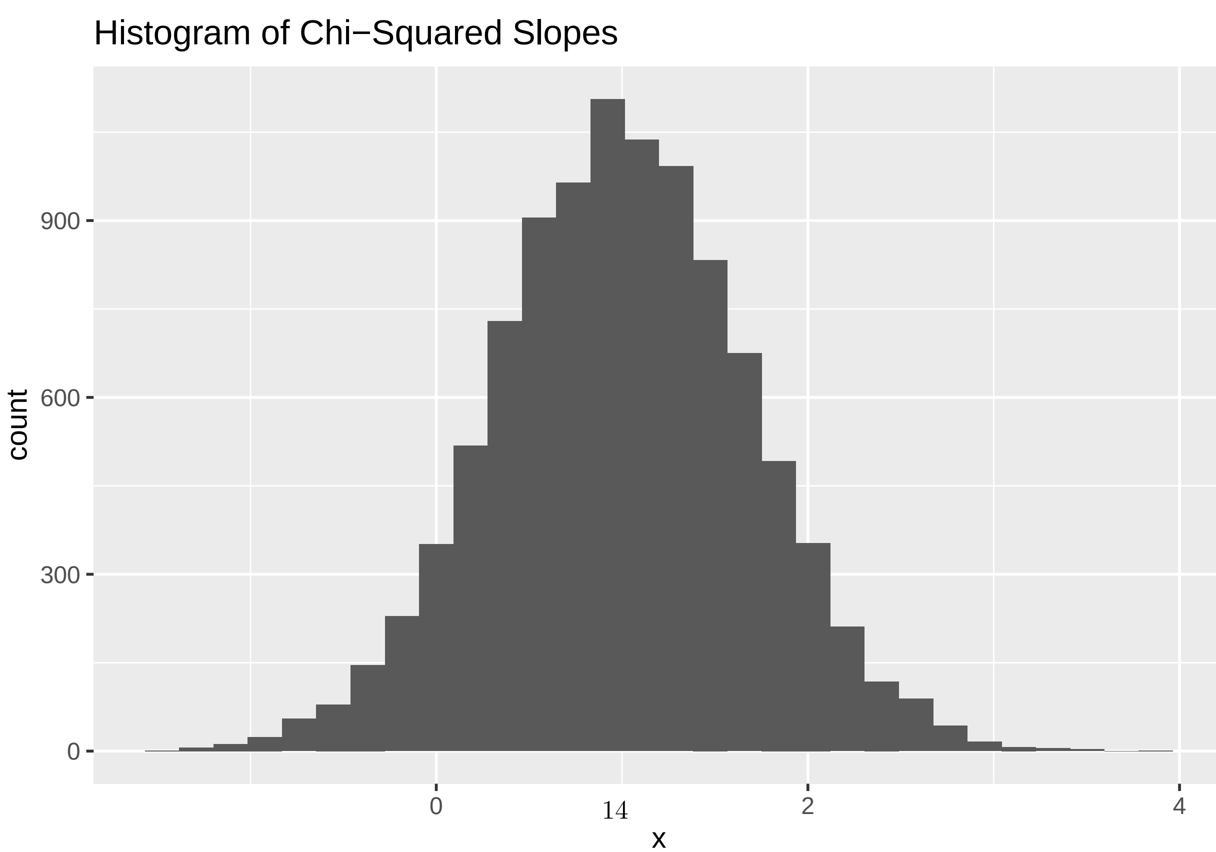
```r
powerdf[nrow(powerdf) + 1 ,] = c(3.0, .994)

ggplot(data = powerdf) +
  geom_point(aes(x = pop_prop, y = power)) +
  geom_line(aes(x = pop_prop, y = power, color = "red")) +
  labs(title = "Plot of Power as a Function of the True Population Proportion of Treatment") +
  theme_bw()
```


Plot of Power as a Function of the True Population Proportion of Treatment
power
1.00
0.75
0.50
0.25
1.5
2.0
2.5
3.0
pop_prop
colour
red

Part 2 Normality

Question 1





Question 2.

```
sd(chidata$x)
```

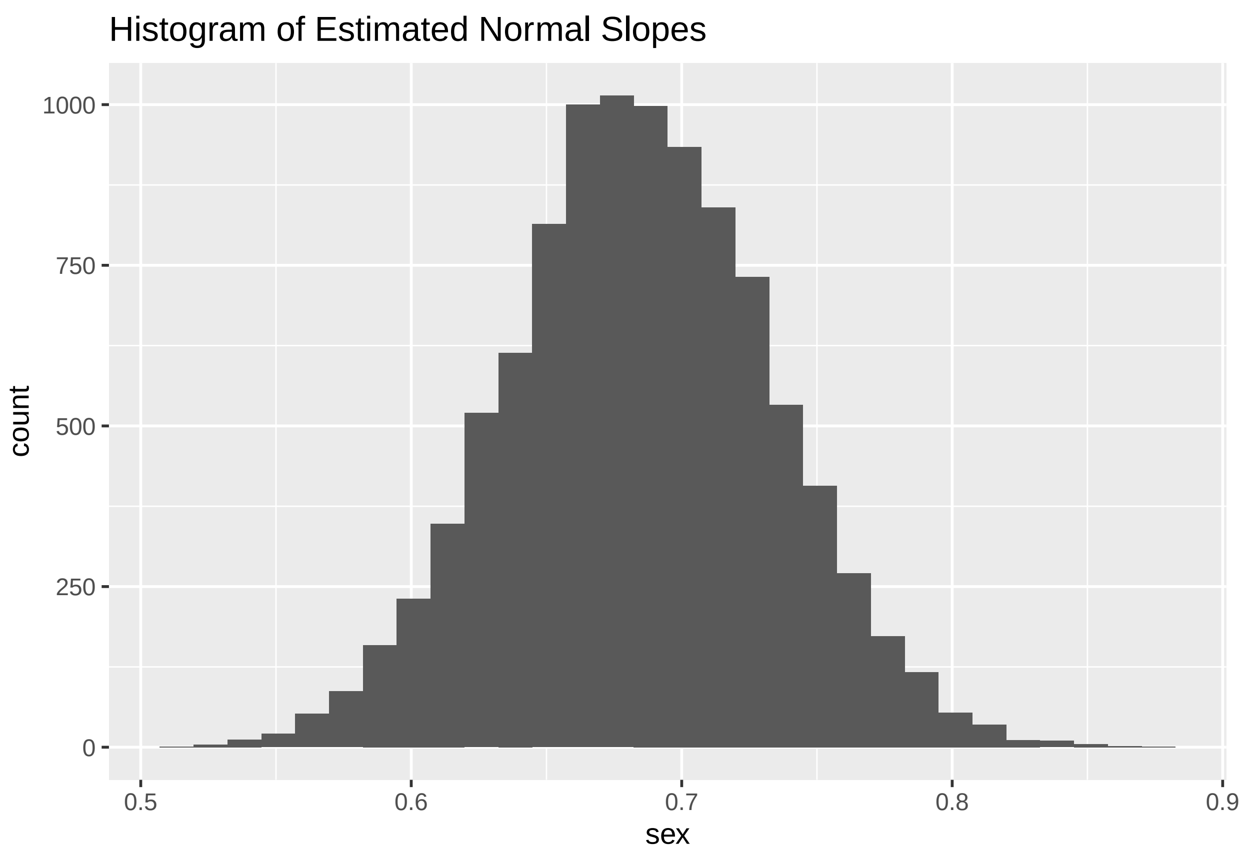
```
## [1] 0.6842273
```

```
sd(normaldata$x)
```

```
## [1] 0.6979592
```

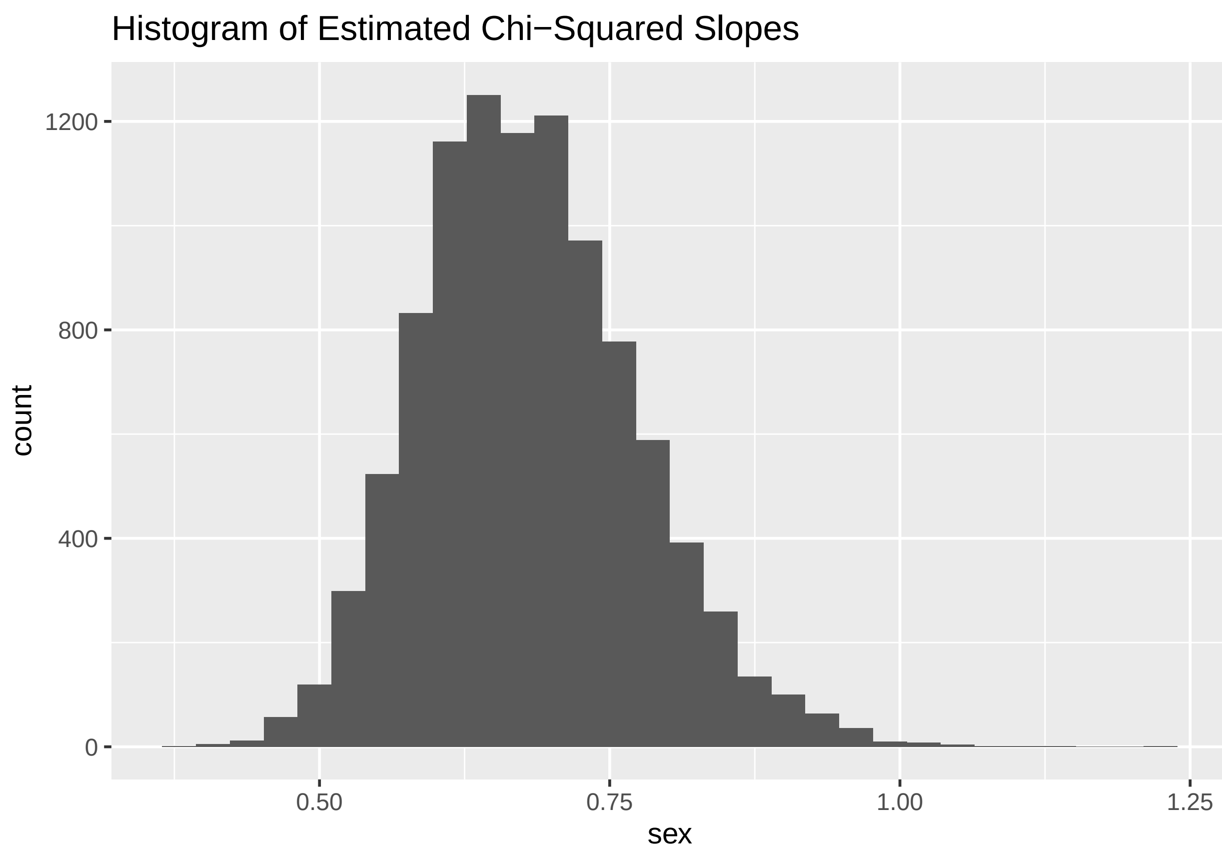
```
ggplot() +
  geom_histogram(data = normaldata, aes(x = sex)) +
  labs(title = "Histogram of Estimated Normal Slopes")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
ggplot() +
  geom_histogram(data = chidata, aes(x = sex)) +
  labs(title = "Histogram of Estimated Chi-Squared Slopes")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Question 3.

```
chidataci <- chidata %>%
  mutate(include_0 = cixlower < 0 & cixupper > 0 )

chidataci %>%
  count(include_0)
```

```
##   include_0    n
## 1     FALSE  505
## 2      TRUE 9495
```

```
normaldataci <- normaldata %>%
  mutate(include_0 = cixlower < 0 & cixupper > 0 )

normaldataci %>%
  count(include_0)
```

```
##   include_0    n
## 1     FALSE  537
## 2      TRUE 9463
```